WORKED SOLUTION

STEP II 2017, Question 8

Pure • Vectors and Orthocentre

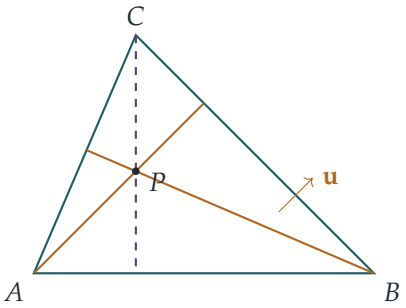
STEP II 2017, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q8

The key idea

This is the vector proof that the three altitudes of a triangle meet at a single point, the *orthocentre*. The trick for finding where two lines $\mathbf{a}+\lambda\mathbf{u}$ and $\mathbf{b}+\mu\mathbf{v}$ meet is to take the scalar product of the equation with a vector perpendicular to $\mathbf{v}$: that kills μ at a stroke. Here such a vector is handed to us, because $\mathbf{v}$ is perpendicular to CA, so $\mathbf{c}-\mathbf{a}$ does the job.



Solution

The first altitude

The line through A perpendicular to BC passes through $\mathbf{a}$ and has direction $\mathbf{u}$ (we are in a plane, so every vector perpendicular to BC is a multiple of $\mathbf{u}$). Its equation is

$$\mathbf{r}=\mathbf{a}+\lambda\mathbf{u}.$$

Similarly the line through B perpendicular to CA is $\mathbf{r}=\mathbf{b}+\mu\mathbf{v}$.

Finding P

These two lines are not parallel: if $\mathbf{u}$ and $\mathbf{v}$ were parallel then BC and CA, being perpendicular to them in the same plane, would be parallel, which is impossible for a triangle. So they meet at exactly one point P, where

$$\mathbf{a} + \lambda\mathbf{u} = \mathbf{b} + \mu\mathbf{v}.$$

Take the scalar product of both sides with $\mathbf{c} - \mathbf{a}$. Since $\mathbf{v}$ is perpendicular to CA, $\mathbf{v} \cdot (\mathbf{c} - \mathbf{a}) = 0$, and μ disappears:

$$\mathbf{a} \cdot (\mathbf{c} - \mathbf{a}) + \lambda\,\mathbf{u} \cdot (\mathbf{c} - \mathbf{a}) = \mathbf{b} \cdot (\mathbf{c} - \mathbf{a}), \qquad \text{so} \qquad \lambda\,\mathbf{u} \cdot (\mathbf{c} - \mathbf{a}) = (\mathbf{b} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a}).$$

Ask yourself Are we allowed to divide by $\mathbf{u} \cdot (\mathbf{c} - \mathbf{a})$?

Yes. If $\mathbf{u} \cdot (\mathbf{c} - \mathbf{a}) = 0$ then $\mathbf{u}$ would be perpendicular to CA as well as to BC; in a plane that forces CA and BC to be parallel, which they are not. Hence

$$\mathbf{p} = \mathbf{a} + \frac{(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a})}{\mathbf{u} \cdot (\mathbf{c} - \mathbf{a})}\,\mathbf{u}.$$

CP is perpendicular to AB

We need $(\mathbf{p} - \mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}) = 0$. Write λ for the coefficient found above. Then

$$(\mathbf{p} - \mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}) = (\mathbf{a} - \mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}) + \lambda\,\mathbf{u} \cdot (\mathbf{b} - \mathbf{a}).$$

The useful observation is that, because $\mathbf{u}$ is perpendicular to BC, $\mathbf{u} \cdot (\mathbf{b} - \mathbf{c}) = 0$, so

$$\mathbf{u} \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{u} \cdot (\mathbf{b} - \mathbf{c}) + \mathbf{u} \cdot (\mathbf{c} - \mathbf{a}) = \mathbf{u} \cdot (\mathbf{c} - \mathbf{a}).$$

Therefore $\lambda\,\mathbf{u} \cdot (\mathbf{b} - \mathbf{a}) = \lambda\,\mathbf{u} \cdot (\mathbf{c} - \mathbf{a}) = (\mathbf{b} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a})$, and

$$(\mathbf{p} - \mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}) = -(\mathbf{c} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a}) + (\mathbf{b} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a}) = 0.$$

Ask yourself Does a zero scalar product really give a *line* CP?

Only if $P \neq C$, and this is where "non-right-angled" is used. If $P = C$ then C lies on the line through A perpendicular to BC, so $AC \perp BC$ and the angle at C is a right angle, which is excluded. So $\mathbf{p} - \mathbf{c} \neq \mathbf{0}$, and since $\mathbf{b} - \mathbf{a} \neq \mathbf{0}$, the zero scalar product says exactly that CP is perpendicular to AB. Thus the three altitudes of the triangle all pass through P.

Checks

Take $A = (0,0)$, $B = (5,0)$, $C = (1.5,3.5)$ and $\mathbf{u} = (1,1)$, which is perpendicular to $\mathbf{c} - \mathbf{b} = (-3.5,3.5)$. Then $(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a}) = 7.5$ and $\mathbf{u} \cdot (\mathbf{c} - \mathbf{a}) = 5$, so $\lambda = 1.5$ and

$P = (1.5, 1.5)$. This lies directly below C, so CP is vertical and perpendicular to AB, and $(\mathbf{p} - \mathbf{b}) \cdot (\mathbf{c} - \mathbf{a}) = (-3.5)(1.5) + (1.5)(3.5) = 0$, so P is on the altitude from B too. The general result was also confirmed by computer on many random triangles. Notice too that the formula for $\mathbf{p}$ is unchanged if $\mathbf{u}$ is replaced by $k\mathbf{u}$, as it must be.

Common slips

- Dotting with $\mathbf{v}$ or $\mathbf{u}$ instead of with $\mathbf{c} - \mathbf{a}$: this leaves both parameters in the equation.
- Dividing by $\mathbf{u} \cdot (\mathbf{c} - \mathbf{a})$ without saying why it is non-zero.
- Missing the step $\mathbf{u} \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{u} \cdot (\mathbf{c} - \mathbf{a})$, which is what makes the final cancellation work.

Beyond STEP

The orthocentre H, centroid G and circumcentre O of any triangle lie on one line, the Euler line, with $\overrightarrow{OH} = 3\overrightarrow{OG}$; with O as origin this is the neat statement $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$, which you can prove with the same scalar-product ideas. Scalar products of this kind are the starting point of the inner-product spaces studied in a first course on linear algebra; Strang's *Introduction to Linear Algebra* is a good place to see them developed.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.